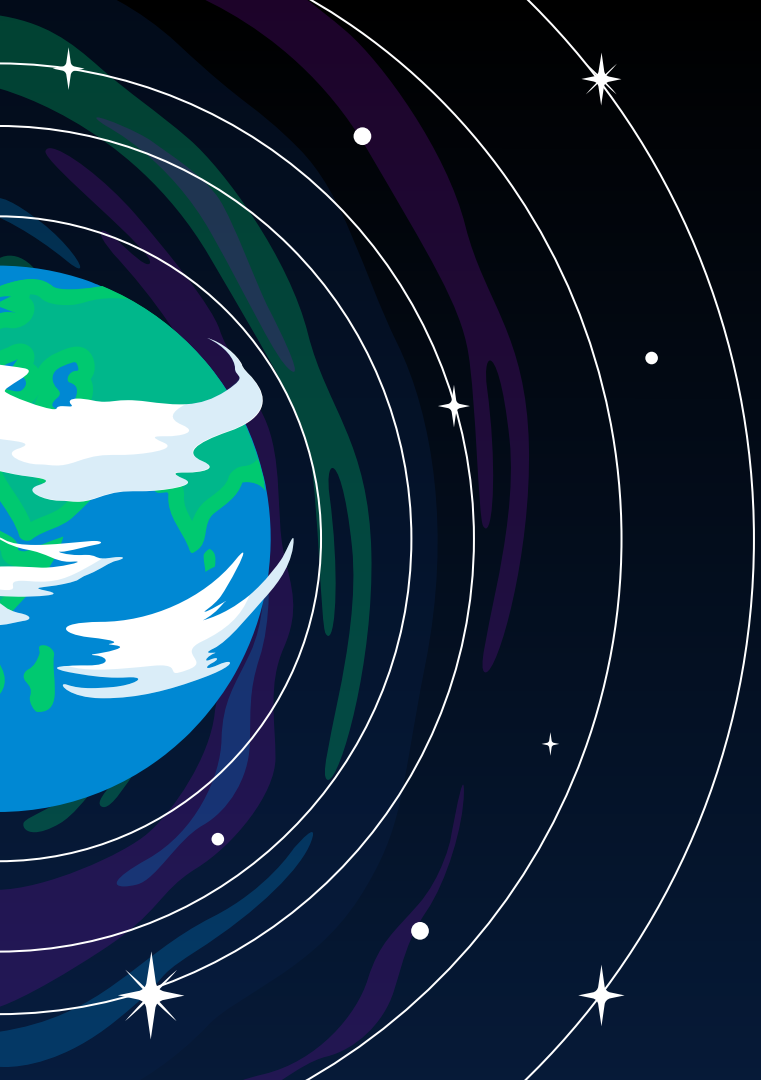


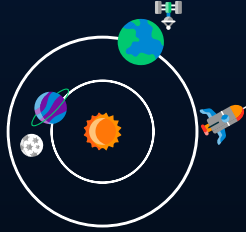


SOLAR SYSTEM SIMULATOR

Foundations of Spatial Computing and
Applications in Augmented and Virtual Reality

Asier Yániz Ibáñez

STATE OF ART



“Dual-Channel: separate processing channels for visual and verbal information”

“People learn more deeply from words and graphics combined than from either alone”

The Cognitive Theory of Multimedia Learning – Richard E. Mayer

STATE OF ART

PROJECT	VISUAL 3D	MISSIONS	CONFIGURATION	ANALYTICS	ONLINE
Academic WebGL Orrery / jsOrrery	✓	✗	✗	✗	✓
Industry NASA Eyes	✓	✓	✗	✗	✓
Industry Stellarium	✓	✗	✗	✗	✗
Industry Celestia / Space Engine	✓	✗	✗	✗	✗
This project SOLAR SYSTEM SIMULATOR	✓	✓	✓	✓	✓

MODULAR STRUCTURE

CORE

Scene · Camera · Renderer · Loop · Controls

SYSTEMS

Solar System · Mission · Interaction · Camera

OBJECTS

Sun · Planet · Natural Satellite · Artificial Satellite · Orbit

UI

Controls · Planet Info · Satellite Info · Mission

DATA

Planets · Natural satellites · Artificial satellites · Analytics

TECHNOLOGY STACK

THREE JS

MOTOR 3D

WebGL rendering

VITE

BUNDLER

Fast development and
optimized builds

ORBIT CONTROLS

CAMERA

Free navigation

DRACO

COMPRESION

Efficient 3D model loading

VANILLA JS

LOGIC

Without additional
frameworks

CSS / HTML

UI

Interface and animation
aesthetics

SOLAR SYSTEM EXPLORATION

PLANETS & ORBITS

- Elliptic orbits
- Axial inclination
- Textures



NATURAL SATELLITES

- Not all the moons
- Orbits & textures

ARTIFICIAL SATELLITES

- 3D models
- Different orbits

INTERACTION & CAMERA

- Zoom & rotation
- Click to follow & see information

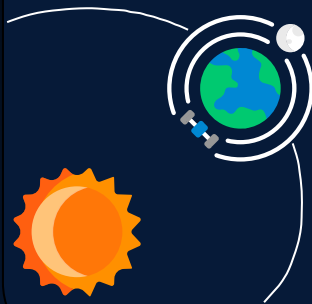
IMPLEMENTATION & CHALLENGES

ORBITS



$$x = \cos(\theta) \cdot a$$
$$z = \sin(\theta) \cdot (a\sqrt{1 - e^2})$$

PIVOT



HITBOX



EXTENSION 1: SIMULATION CONFIGURATION



Show or not

Moons · Orbits ·
Satellites



Speed

Time scale



Illumination

Add ambient
illumination



Pause

Stop orbit
movement



Mission mode

Specific
configuration

IMPLEMENTATION & CHALLENGES

MISSION MODE

Freeze orbits
position

Hide all objects
unrelated to the
mission

Deactivate Orbit
Controls

ILLUMINATION

Sun source of all
lighting



Add ambient light

EXTENSION 2: MISSION SIMULATOR

ARTEMIS II

Lunar fly over

NASA

4 astronauts

APRIL 2026

10 days



Phases:

- Launch
- Parking orbit
- Trans-lunar injection
- Translunar coast
- Lunar flyby
- Free return
- Re-entry
- Splashdown

IMPLEMENTATION & CHALLENGES

TRAJECTORY

Catmull-Rom

Curve A & Curve B

Phase $\rightarrow [a, b]$

PARTICLES

Points material

290 random particles

Combustion active

EXTENSION 3: METRICS SYSTEM



Student



Researcher

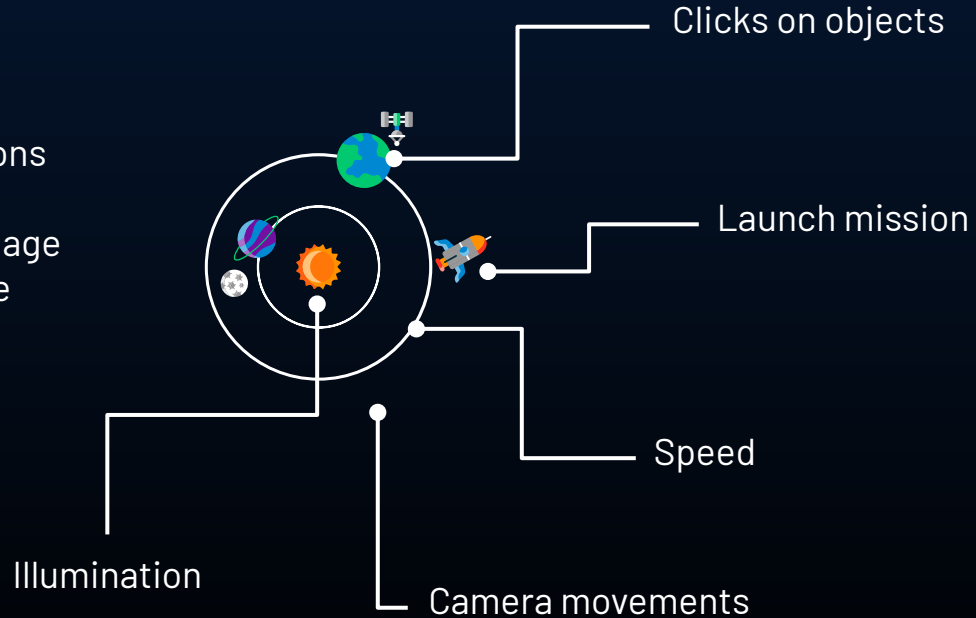


Enthusiast



Admin

- Users
- Sessions
- Time
- Language
- Device



IMPLEMENTATION & CHALLENGES

LOCAL STORAGE

beforeUnload -> Save session info

localStorage -> Browser storage

Browser / Port / Domain

DEMO



CONCLUSIONS

01 THREE JS

Enables high-quality 3D rendering without plugins, applications, or specialized hardware.

03 COMPLETE APP

No complete application in this field that offers everything mine does

02 TRADE OFF

Trade-off between accuracy and usability depending on the app goal

04 UNDERSTANDING PURPOSE

The app helps improve understanding of the Solar System

FUTURE WORK

01

Replace localStorage with a real database and backend

02

Add more mission simulations

03

Explore a WebXR mode for VR headsets

04

Develop the web depending on the user types



The background is a deep black space filled with numerous small white stars of varying sizes. A bright, glowing white star is positioned in the upper center. On the right side, the curved horizon of the Earth is visible, showing a blue atmosphere and a dark, cratered surface. Abstract, wavy shapes in shades of green and purple are layered along the left and right edges of the image. In the bottom left corner, the text "THANKS FOR YOUR ATTENTION" is written in a bold, white, sans-serif font, arranged in two lines.

**THANKS FOR YOUR
ATTENTION**